Discrete Probability Distributions

1. State what kind of distribution is

   (a) $f(x)=1/N$ where $x=x_1,x_2,\ldots,x_N$

   (b) $b(x;n,p)=\binom{n}{x}p^x q^{n-x}$ where $x=0,1,2,\ldots,n$

   (c) $f(x)=\dfrac{\lambda^x e^{-\lambda}}{x!}$ where $x=0,1,2,\ldots$

2. When a child is exposed to a certain contagious disease, the probability is 0.4 that he/she will show symptoms. A class of children have all been exposed to the disease and are monitored for symptoms.

   (a) What is the probability that the fourth child monitored will be the first to show symptoms?

   (b) What is the expected number of children monitored before the first is found to show symptoms?

   (c) What is the probability that the tenth child monitored will be the third to show symptoms?

   (d) What is the expected number of children monitored before three are found to show symptoms?

3. A large chain retailer purchases a certain kind of electronic devices from a manufacturer. The manufacturer indicates that the defective rate of the device is 3%.

   (a) The inspector of the retailer randomly picks 20 items form the shipment. What is the probability that there will be at least one defective item among this 20?

(b) Suppose that the retailer receives 10 shipments in a month and the inspector randomly tests 20 devices per shipment. What is the probability that there will be 3 shipments containing at least one defective device?

4. At “peak time” during the university clearance period a telephone exchange is very near capacity, so callers have difficulty placing their calls. It may be of interest to know the number of attempts necessary in order to gain a connection. Let $p=0.05$ be the probability of a connection during busy time.
   (a) We are interested in knowing the probability that 5 attempts are necessary for a successful call.
   (b) What is the expected number of calls to make a connection?
5. Suppose you have a fair six-sided die. Each face of the die is equally likely to land face up.
   (a) What is the probability of rolling a 3 on a single roll of the die?
   (b) If you roll the die twice, what is the probability of getting a sum of 7?
6. A call center receives an average of 4 customer service calls per hour.
   (a) What is the probability that the call center receives exactly 2 calls in the next hour?
   (b) On average, how many calls would you expect the call center to receive in a 30-minute period?
   (c) What is the probability of receiving at least 5 calls in a given hour?
7. Consider a fair six-sided die.
   (a) Simulate the process of rolling the die 100 times using R. Write an R script to generate a sequence of 100 random outcomes, each representing the result of a single roll of the die.

(b) Calculate the empirical probabilities (frequencies) of each outcome (1 through 6) based on the simulation. Compare these empirical probabilities with the expected theoretical probabilities for a fair six-sided die.

(c) Visualise the results using a bar chart, where the x-axis represents the outcomes (1 through 6) and the y-axis represents the empirical probabilities.

(d) Discuss any observations or discrepancies between the empirical and theoretical probabilities and explain how increasing the number of simulations might affect the results.

8. In R, attempt the following questions and visualise the scenario. Out of 800 families with 5 children, how many would you expect to have

   (a) 5 boys

   (b) 2 girls

   (c) Either 2 or 3 girls